Quantum Fault Tolerance Meets Toom's Contours

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1 Toric Encoded Circuits

Definition 1.1 (Troic Encoded Circuit)**.** Let C be a quantum circuit. We say that C is a *Toric encoded circuit* if the following holds:

1. There is a Toric code $\mathcal{C}$ such that the qubits of $\mathcal{C}$ can be partitioned into registers $R_1, \ldots, R_k$, and, at any time, each of the registers is encoded either by the *nice encoding of* $\mathcal{C}$ or by the *nice encoding of* $\phi \circ H(\mathcal{C})$[1]. At time $t = 0$, the registers are initialized to be either $|\bar{0}\rangle$ or encoded $|\mathbf{S}\rangle$.

2. The layers of C at even times are sweep-cycles, and the layers at odd times are realizations of one of the logical $\mathbf{H}, \mathbf{S}, \mathbf{CX}, \mathbf{X}$, or $\mathbf{Z}$ gates.

Definition 1.2 (The Boundery of Deviation ∂D_t)**.** Let C be a Toric encoded circuit, with registers $R_1, .., R_l$ and let $\mathcal{E}$ be a set of faults. Consider the subjection $C[\mathcal{E}]$, For a time t, let $A_t, B_t \subset \{R_1, .., R_l\}$ be the registers that encoded, at time t in $C[\mathcal{E}]$, by $\mathcal{C}$ and $\phi \circ \mathbf{H}$ respectively. Denote by $\mathcal{D}_t$ the deviation in $C[\mathcal{E}]$ at time t. We define the *X-type, and the Z-type boundaries* of $\mathcal{D}_t$, denoted by ∂D_t^X and ∂D_t^Z, as follow:

$$\partial \mathcal{D}_t^X := \bigcup_{R_i \in A_t} \partial \left| R_{i,t} \right\rangle \cup \bigcup_{R_i \in B_t} \partial \left(\phi \circ \mathbf{H} \right) \left| R_{i,t} \right\rangle$$
$$\partial \mathcal{D}_t^Z := \bigcup_{R_i \in B_t} \partial \left| R_{i,t} \right\rangle \cup \bigcup_{R_i \in A_t} \partial \left(\phi \circ \mathbf{H} \right) \left| R_{i,t} \right\rangle$$

Definition 1.3 (Base graph G_o)**.** Let C be a quantum circuit, at depth d, with registers $R_1, \ldots, R_k$, all of which are encoded by Toric codes with the same parameters (i.e., dimension, length, etc.). Let's denote by $\mathcal{G}_1, .., \mathcal{G}_k$ the Toric complexes that induce the codes. We define the base graph of C, denoted by $G_o = (V_o, E_o)$ as follows:

[1] Notice that although the code space is invariant to the application of $\phi \circ \mathbf{H}$, the positive direction of the underlying Toric complex is flipped.

1. The vertices V_o are tuples, where the first coordinates correspond to a vertex in the union over the vertices in $\mathcal{G}_1, \ldots, \mathcal{G}_k$, and the second coordinate is associated with a time. Namely:

$$V_o := \bigcup_i V(\mathcal{G}_i) \times [d]$$

2. The edges E_o are derived from the original edges in $\mathcal{G}_1, \ldots, \mathcal{G}_k$, with the addition that any two vertices in preceding times t and $t+1$ that support qubits (faces) such that C acts non-trivially on those qubits at time t are also connected by an edge. Namely:

$$\begin{aligned}
E_o := \Big\{ &\{(v,t),(u,t')\} : \text{ if} \\
&\text{either there exists } i \text{ such } \{v,u\} \in \mathcal{G}_i \text{ and } t' = t, t+1 \\
&\text{or there exists } u_i = (g_i, l_i) \in C \text{ such } v, u \in l_{i,2} \text{ and } t = l_{i,1}, t' = l_{i,1} + 1 \Big\}
\end{aligned}$$

[COMMENT] Here we should add that a vertex v belongs to location l if there is a qubit f that v belongs to its associated face.

Definition 1.4 (Tooms Contour Graph of $C(\mathcal{E})$)**.** For a quantum circuit C and set of faults $\mathcal{E}$, let us denote the base graph of $C[\mathcal{E}]$ by G_o. For each time t, denote the deviation induced by $\mathcal{E}$ at time t by $\mathcal{D}_t$, so $\partial \mathcal{D}_t$ is the syndrome that the qubits in the deviation admit. We define the *Tooms contour graph* $\mathcal{H} = (V_{\mathcal{H}}, E_{\mathcal{H}})$ as follows:

1. The vertices $V_{\mathcal{H}}$ are partitioned into two types, X-type vertices $V_{\mathcal{H},X}$ and Z-type vertices $V_{\mathcal{H},Z}$. Each of the X/Z-type corresponds to the vertices supporting the X/Z-boundary of $\mathcal{D}$. Namely:

$$\begin{aligned}
V_{\mathcal{H},X} &:= \Big\{ v_t : v_t = (v,t) \in V_o \text{ and there exists a face } f \in \partial \mathcal{D}_t^X \text{ such } v \in f \Big\} \\
V_{\mathcal{H},Z} &:= \Big\{ v_t : v_t = (v,t) \in V_o \text{ and there exists a face } f \in \partial \mathcal{D}_t^Z \text{ such } v \in f \Big\}
\end{aligned}$$

Similarly, the Z-type vertices $V_{\mathcal{H},Z}$ are the vertices $(v,t) \in V_o$, that satisfy that v belongs to a face associated with a check, which, at time t, belongs to the $(\partial D_t)|_Z$.

2. The edges $E_{\mathcal{H}}$ is partitioned into two types $A_{\mathcal{H}}$ and $F_{\mathcal{H}}$, where for any physical gate of $(g, l = (t, S))$ of $C[\mathcal{E}]$ we add edge to $A_{\mathcal{H}}$ as follow:

   (a) If g is physical **CX** between faces f_1 into f_2 at time t then we add:

$$\begin{aligned}
&\Big\{ \{(u,t),(v,t+1)\} : u \in f_1, v \in f_1 \cup f_2 \text{ and } v, u \in V_{\mathcal{H},X} \Big\} \\
&\Big\{ \{(u,t),(v,t+1)\} : u \in f_2, v \in f_1 \cup f_2 \text{ and } v, u \in V_{\mathcal{H},Z} \Big\}
\end{aligned}$$

   (b) If g is physical $\phi \circ \mathbf{H}$.

$$\begin{aligned}
A_{\mathcal{H}} &= \Big\{ e = \{u,v\} \in E_o : u, v \in V_{\mathcal{H}} \text{ and } T(v) = T(u) \pm 1 \Big\} \\
F_{\mathcal{H}} &= \Big\{ e = \{u,v\} \in E_o : \text{ exists } w \in V_{\mathcal{H}} \text{ such } \{v,w\}, \{u,w\} \in A_{\mathcal{H}} \Big\}
\end{aligned}$$

By definition of the construction, $G_{\mathcal{H}}$ is a time-graph.

Definition 1.5 (Time Graph of $C(\mathcal{E})$)**.** For a quantum circuit C and set of faults $\mathcal{E}$, let us denote the base graph of $C[\mathcal{E}]$ by G_o. For each time t, denote the deviation induced by $\mathcal{E}$ at time t by $\mathcal{D}_t$, so $\partial\mathcal{D}_t$ is the syndrome that the qubits in the deviation admit, denote by $\partial\mathcal{D}_t^X$ and by $\partial\mathcal{D}_t^Z$ the $X-Z$ decomposition of $\partial\mathcal{D}_t$. We define the history graph $\mathcal{H} = (V_{\mathcal{H},X} \cup V_{\mathcal{H},Z}, E_{\mathcal{H}})$, in recursive manner as follow:

1. For the empty circuit. The vertices in $V_{\mathcal{H},X}$ are the vertices of $V_o|_1$ that support ∂D_t^X. Similarly, the vertices of $V_{\mathcal{H},\mathcal{Z}}$ are the vertices of $V_o|_1$ that support ∂D_t^Z.
2. Let $\mathcal{H}$ be the history graph of $C[\mathcal{E}]$ restricted to the first $d-1$ layers. Let U be the d-layer, then define the history graph corresponding to the first d layers of $C[\mathcal{E}]$ to be obtained by adding vertices and edges to $\mathcal{H}$ as follows:
   (a) If U is a noise gate, namely $U \subset \mathcal{E}$, then denote by $\mathcal{E}_t^X$ and $\mathcal{E}_t^Z$ the X, Z decomposition of $\mathcal{E}_t$. Then, for any vertex in $\mathcal{G}$
   (b) Else, denote by $U_1, .., U_l$ the gates in U. If U_1 is a $\mathbf{CX}^{\otimes n}$ gate acting on registers R_1 into R_2, then for each vertex $v \in V_{\mathcal{H},X}|_{R_1,t=d-1}$ add the counter parts of v in R_1,R_2 at time $t = d+1$ into $V_{\mathcal{H},X}$ and connect them by arrows to v. Similarly, for any vertex $v \in V_{\mathcal{H},Z}|_{R_2,t=d-1}$ add the counter parts of v in R_2,R_2 at time $t = d+1$ into $V_{\mathcal{H},X}$ and connect them by arrows to v.

[COMMENT] What we have to do is take the first definition, but split the vertex into X-type and Z-type. Then we connect the boundary of two registers in time if the gate applied is I or $\mathbf{CX}$; otherwise, in case we apply Hadamard, we cross the type. If we apply a Sweep cycle, then again we connect only vertices of the same type.